UNIVERSITY NAME

Rigid Body Mechanics — Make-Up Exam

Year: 2015

Exercise 1:

Consider a perfect rigid body S composed of a homogeneous disk S_1 of center D, radius a, and mass M, and a homogeneous rod S_2 of center of mass T, length L, and mass m, welded to the disk at point D.

The system is in motion in the vertical plane X_0OY_0 of the reference frame $R_0(O, X_0, Y_0, Z_0)$, with basis $(\bar{x}_0, \bar{y}_0, \bar{z}_0)$, assumed to be fixed and Galilean. The disk S_1 rolls without slipping on the axis OY_0. I denotes the geometric contact point of S_1 with the axis OY_0, and y represents the coordinate of point D on the axis OY_0. The position of the rod relative to the vertical is defined by the angle θ. We denote by $\bar{R}$ the reaction force exerted on the system at point I.

1. (a) State the kinetic energy theorem.
   (b) Prove this theorem.
2. Calculate the inertia matrix at point D of the disk S_1.
3. Calculate the inertia matrix at point T of the rod S_2 and deduce its inertia matrix at point D by applying Huygens' theorem.
4. Calculate the sliding velocity of the solid and deduce the condition for rolling without slipping.
5. Determine, as a function of the angle θ, the absolute velocity of point T, expressed in the basis of R_0.
6. Calculate the kinetic energy of the system $E_c(S/R_0)$ as a function of θ and $\dot{\theta}$.
7. (a) By applying the kinetic energy theorem to the system in R_0, determine the first integral of energy.
   (b) Deduce the differential equation of motion of the system.
8. Limiting to the case of small angles and keeping only first-order terms in θ and $\dot{\theta}$, specify the nature of the motion and calculate the period of small oscillations T_0.
9. Retrieve the differential equation of motion of the system by applying the angular momentum theorem at the geometric point I in the reference frame R_0.

Answer Area

1. (a)
   (b)
2.
3.
4.
5.
6.
7. (a)
   (b)
8.
9.